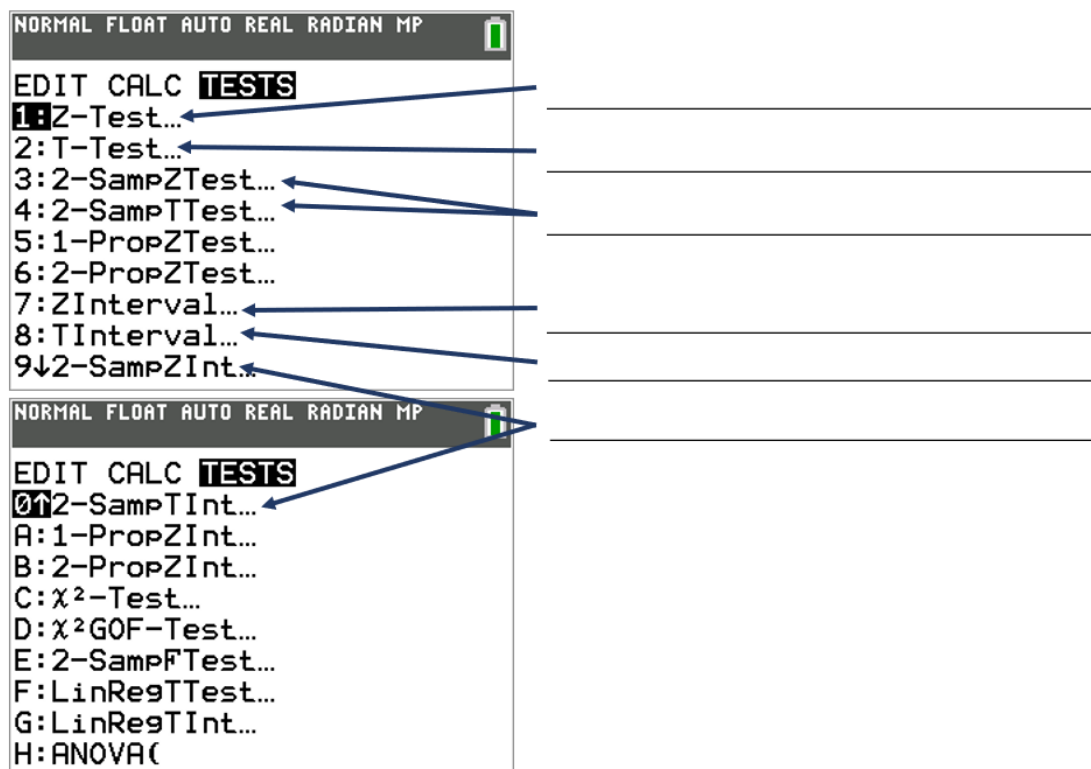


TI-84 Calculator Commands for Inference



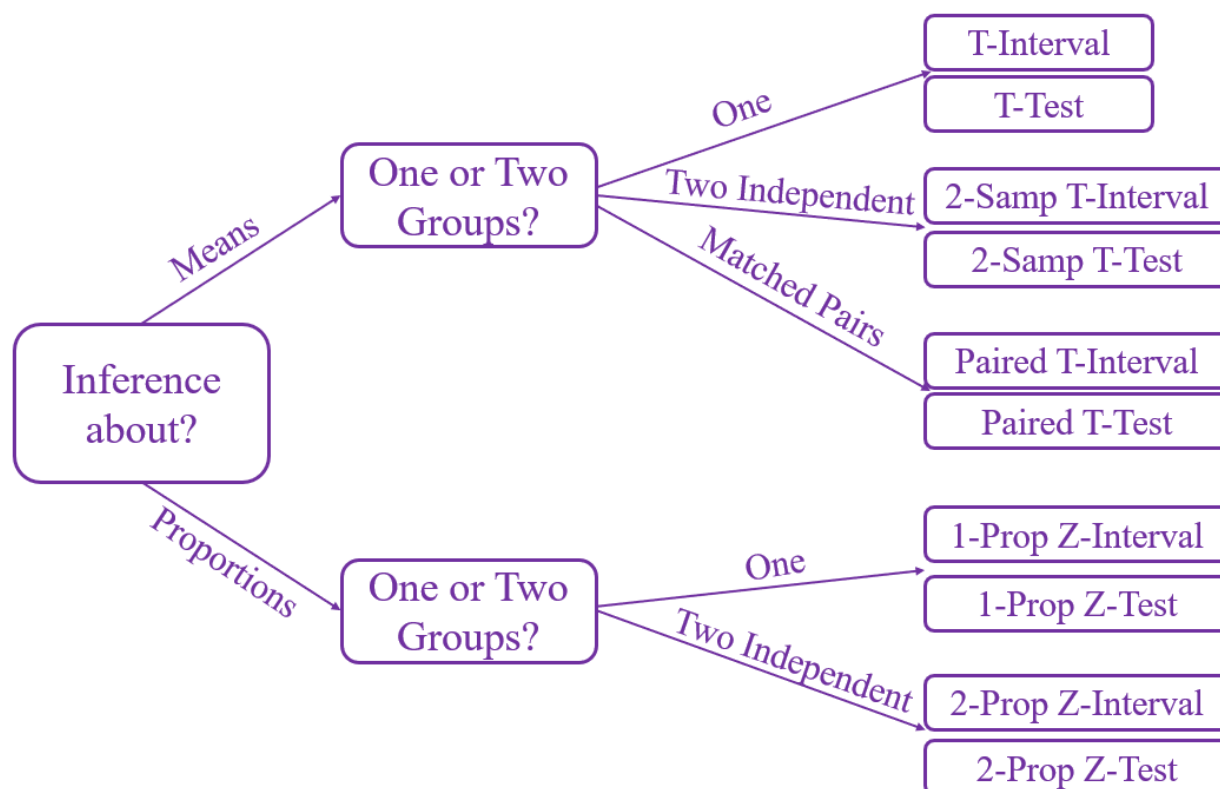
- There are 8 total inference options for means on your TI-84.
- For each inference method, you can use data that you have entered into your lists OR you can use statistics that are given in the problem.
- Because the Z inference procedures for means involve the unique circumstance where you have to know σ , we will *most likely* not be using those.
- Special Note for Two Sample Inference for Means
 - DO NOT POOL FOR MEANS
 - When you run your significance test, your calculator will give you an exact value for your df

Let's complete the table for the 4 common T procedures:

<i>Inference Method</i>	<i>Formula</i>	<i>Input</i>	<i>Output</i>
One Sample T Interval for Means		8: Tinterval 	

Two Sample T Interval for Means		0: 2-SampTInt NORMAL FLOAT AUTO REAL RADIAN MP 2-SampTInt Inpt:Data Stats $\bar{x}_1$:15 Sx_1 :2.566 n_1 :35 $\bar{x}_2$:18 Sx_2 :1.245 n_2 :40 C-Level:0.95 ↓Pooled:No Yes	NORMAL FLOAT AUTO REAL RADIAN MP 2-SampTInt (-3.958, -2.042) df=47.68563164 $\bar{x}_1$:15 $\bar{x}_2$:18 Sx_1 :2.566 Sx_2 :1.245 n_1 :35 n_2 :40
One Sample T Test for Means		2: T-Test NORMAL FLOAT AUTO REAL RADIAN MP T-Test Inpt:Data Stats μ_0 :13.5 $\bar{x}$:15 Sx :2.566 n :35 μ : $\neq \mu_0$ $< \mu_0$ $> \mu_0$ Color: BLUE Calculate Draw	NORMAL FLOAT AUTO REAL RADIAN MP T-Test $\mu \neq 13.5$ t=3.458347496 p=0.0014808508 $\bar{x}$:15 Sx :2.566 n=35
Two Sample T Test for Means		4: 2-SampTTest NORMAL FLOAT AUTO REAL RADIAN MP 2-SampTTest Inpt:Data Stats $\bar{x}_1$:15 Sx_1 :2.566 n_1 :35 $\bar{x}_2$:18 Sx_2 :1.245 n_2 :40 μ_1 : $\neq \mu_2$ $< \mu_2$ $> \mu_2$ ↓Pooled:No Yes	NORMAL FLOAT AUTO REAL RADIAN MP 2-SampTTest $\mu_1 \neq \mu_2$ t=-6.298365433 p=8.987531934E-8 df=47.68563164 $\bar{x}_1$:15 $\bar{x}_2$:18 Sx_1 :2.566 ↓ Sx_2 :1.245

Choosing Your Inference Method



Example

For the following problems, identify which inference method you would use to answer the question.

A. One Sample T Test	B. One Sample T Interval	C. Two Sample T Test	D. Two Sample T Interval	E. Paired T Test
F. Paired T Interval	G. One Proportion Z Test	H. One Proportion Z Interval	I. Two Proportion Z Test	J. Two Proportion Z Interval

_____ Researchers want to estimate the difference in average systolic blood pressure between men and women and provide a 95% confidence interval for this difference.

_____ A surveyor wants to estimate the proportion of customers who are satisfied with their service and provide a 95% confidence interval for this proportion.

_____ A researcher wants to estimate the mean cholesterol level in a population and calculate a 95% confidence interval for the mean cholesterol level based on a sample.

_____ An economist wants to estimate the difference in proportions of two different regions that support a tax increase and provide a 95% confidence interval for the difference.

_____ A fitness coach wants to determine if a new training program significantly changes the running times of athletes by comparing their times before and after the program.

_____ A psychologist wants to estimate the mean difference in stress levels before and after a mindfulness course within the same group of participants and provide a 95% confidence interval for the mean difference.

_____ A teacher wants to know if the average test score of her class is different from the national average of 75.

_____ A medical researcher wants to compare the proportion of patients who respond to two different treatments to determine if there is a significant difference between the treatments.

_____ A company wants to compare the mean productivity of two different departments to see if there is a significant difference.

_____ A political analyst wants to know if the proportion of people who support a new policy is different from 50%.